Isobel Voysey

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Education

2020 – 2024 **PhD, University of Edinburgh & Heriot-Watt University, UK** in Robotics & Autonomous Systems
Thesis title: *A zoomorphic robot for animal welfare education*
Supervised by: Dr Michael Herrmann, Prof. Lynne Baillie, Prof. Joanne Williams

2016 – 2020 **MEng & BA, University of Cambridge, UK** in Engineering
Awarded First Class degree
Thesis title: *Autonomous dishwasher loading from cluttered trays using pre-trained deep neural networks*
Supervised by: Prof. Fumiya Iida

2018 – 2019 **Exchange year, National University of Singapore, Singapore** in Engineering
GPA: 4.55/5.0, equivalent to First Class degree

Research Publications

Journal Articles

J1 **Voysey, I.**, George Thuruthel, T., & Iida, F. (2021). Autonomous dishwasher loading from cluttered trays using pre-trained deep neural networks. *Engineering Reports*, *3*(5), e12321.

Peer-Reviewed Full Paper Conference Contributions

C2 **Voysey, I.**, Baillie, L., Williams, J., & Herrmann, J. M. (2022). Influence of animallike affective non-verbal behavior on children's perceptions of a zoomorphic robot. In *2022 31st IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)* (pp. 1443–1450). IEEE.

C1 **Voysey, I.**, Hughes, J., Thuruthel, T. G., Gilday, K., & Iida, F. (2020). Real world Bayesian optimization using robots to clean liquid spills. In *Annual conference towards autonomous robotic systems* (pp. 196–208). Springer.

Peer-Reviewed Short Paper & Workshop Conference Contributions

W5 Stals, S., **Voysey, I.**, & Baillie, L. (forthcoming). Let's make this fun!: Activities to motivate children and teens to complete questionnaires. In *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*.

W4 **Voysey, I.** (forthcoming). Developing a zoomorphic robot for animal welfare education. In *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*.

W3 **Voysey, I.**, Bettosi, C., Nault, E., Stals, S., & Baillie, L. (2023). Introducing children and young people with sight loss to social robots: A preliminary workshop. In *Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction* (pp. 384–388). **Best LBR Award Winner**.

W2 **Voysey, I.** (2022). Participatory design for the development of a robot for animal welfare education. In *2022 31st IEEE International Conference on Robot and Human Interactive Communication (RO-MAN) Workshop on Participatory Design in Human-Robot Interaction*. IEEE.

W1 Zhou, M., **Voysey, I.**, & Herrmann, J. M. (2022). Is machine learning enough to train robotic pets? In *2022 31st IEEE International Conference on Robot and Human Interactive Communication (RO-MAN) Workshop on Machine Learning for HRI: Bridging the Gap between Action and Perception (ML-HRI)*. IEEE.

In Preparation/Under Review

2 **Voysey**, **I.**, Baillie, L., Williams, J., & Herrmann, J. M. (submitted). Co-designing zoomorphic robot concepts for animal welfare education. *Submitted to International Journal of Social Robotics*.

1 **Voysey**, **I.**, Baillie, L., Williams, J., & Herrmann, J. M. (submitted). The problem of cuteness in zoomorphic robots. *Submitted to ACM Transactions on Human-Robot Interaction*.

Teaching Experience

Spring 2024 — **INFR09032 System Design Project**, University of Edinburgh
Course: 3rd year undergraduate course on design of assistive robotics
Role: Lead teaching assistant
Activities: Acting as the 'client' for 10 groups, assessing monthly group presentations and reports (providing both formative and summative assessment), delivering workshops

Spring 2023, 2024 — **The Scholars Programme**, The Brilliant Club
Course: Self-designed 8-week course on human-robot interaction for 13–18-year-olds from groups underrepresented at university
Role: Teacher
Activities: Designing course and materials (including a handbook, tutorial materials, and assessment materials), delivering weekly tutorials, providing feedback on assignments, supporting students to complete assignments

Spring 2023 — **INFR09032 System Design Project**, University of Edinburgh
Course: 3rd year undergraduate course on design of assistive robotics
Role: Teaching assistant
Activities: Timetabling and organising demonstration days, communicating with students, marking monthly group presentations, marking and giving feedback on individual reflections, delivering workshops

Autumn 2022, 2023 — **INFR10022 Professional Issues**, University of Edinburgh
Course: 3rd year undergraduate course on professional and ethical behaviour in computing and technology development
Role: Tutor
Activities: Leading biweekly seminar discussions, assessing student essays

INFR11130 Machine Learning and Pattern Recognition, University of Edinburgh
Course: 4th year undergraduate course on machine learning
Role: Marker
Activities: Marking courseworks and exam

Spring 2022 — **INFR09032 System Design Project**, University of Edinburgh
Course: 3rd year undergraduate course on design of assistive robotics
Role: PhD expert
Activities: Providing guidance to groups on human-robot interaction during weekly office hour, marking monthly group presentations, marking and giving feedback on individual reflections

Honors & Awards

2024 — **HRI Pioneer**, 19th HRI Pioneers Workshop, HRI 2024
Selected to attend workshop to foster collaboration between distinguished PhD researchers

Honors & Awards (continued)

2023 **Best Late-Breaking Report Award**, ACM/IEEE International Conference on Human-Robot Interaction

2022 **Best Student Spotlight Talk**, CDT-RAS Annual Conference, Edinburgh Centre for Robotics

2020 **Engineering Essay Prize**, Trinity College, University of Cambridge

2016 – 2018 **Samwaies and Blundell Awards for Academic Excellence**, Trinity College, University of Cambridge

Scholarships

2020 – 2024 **EPSRC Doctoral Studentship**, CDT-RAS, Edinburgh Centre for Robotics

2018 – 2019 **Shell Engineering Exchange Scholarship**, University of Cambridge

Academic Service

2024 **Student Volunteer**, ACM/IEEE International Conference on Human-Robot Interaction

2022 – 2024 **Reviewer**, ACM/IEEE International Conference on Human-Robot Interaction 2023–2024, ACM Designing Interactive Systems 2024, IEEE International Conference on Robot and Human Interactive Communication 2022

Public Engagement & Outreach

2021 – 2023 **STEM Ambassador**, STEM Ambassadors Scotland
Speaker and demonstrator, school visits with robots to encourage children to study STEM

2021 – 2022 **Informatics Circle**, School of Informatics, University of Edinburgh
Designer and presenter, computer science activities and newsletters for 7–14-year-olds

2022 **Robot Lab Live**, UK Festival of Robotics, UK-RAS
Presenter, livestreamed demonstration of soft robot designed in conjunction with other PhD researchers at Edinburgh Centre for Robotics

2019 – 2020 **STIMULUS**, Department of Engineering, University of Cambridge
Teaching assistant, A-Level pure and applied maths classes at local sixth form college

College Ambassador, Trinity College, University of Cambridge
Helper, Women in STEM residential for teenagers considering studying STEM subjects at university